EX. NO: 5 — RAIL FENCE CIPHER

Aim:

To implement the Rail Fence transposition technique using C.

Algorithm:

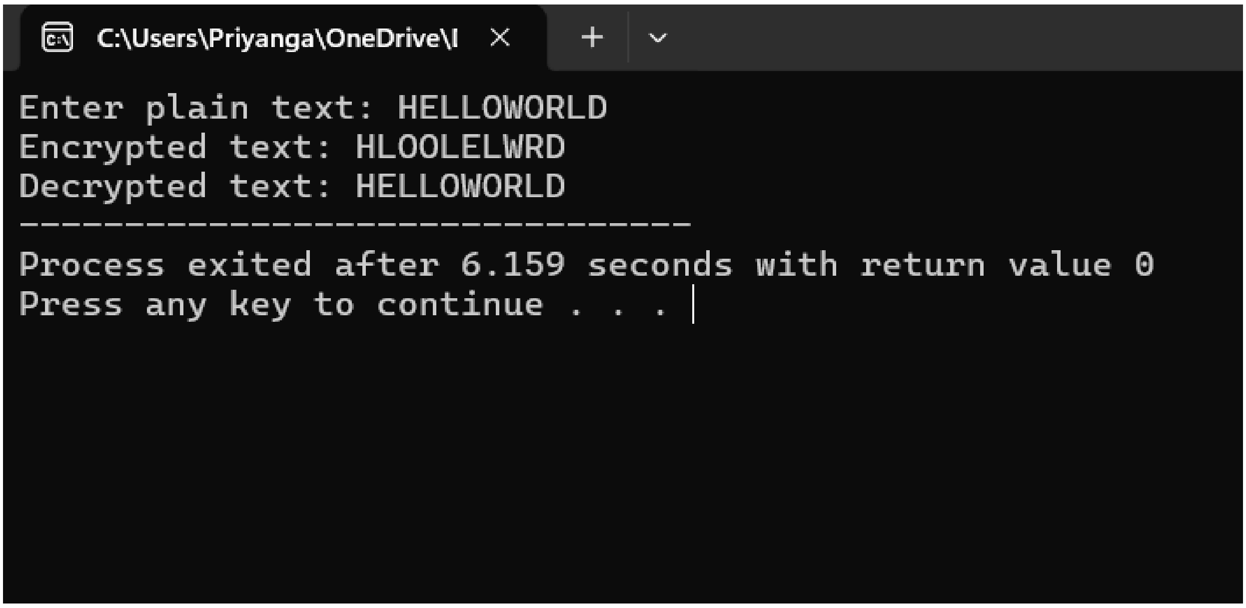
1. Read plaintext.
2. Separate characters into odd and even positions.
3. Combine them to form ciphertext.
4. Reverse the process for decryption.
5. Display plaintext and ciphertext.

Program:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char text[100], cipher[100], decrypt[100];
    int i, j, k, len;
    printf("Enter plain text: ");
    scanf("%s", text);
    len = strlen(text);
    /* Encryption */
    j = 0;
    for(i = 0; i < len; i += 2)
        cipher[j++] = text[i];
    for(i = 1; i < len; i += 2)
        cipher[j++] = text[i];
    cipher[j] = '\0';
    printf("Encrypted text: %s", cipher);
    /* Decryption */
```

```
    k = (len + 1) / 2;
    j = 0;
    for(i = 0; i < k; i++)
    {
        decrypt[j] = cipher[i];
        j += 2;
    }
    j = 1;
    for(i = k; i < len; i++)
    {
        decrypt[j] = cipher[i];
        j += 2;
    }
    decrypt[len] = '\0';
    printf("\nDecrypted text: %s", decrypt);
    return 0;
}
```

Output:



Result:

Thus, the Rail Fence transposition technique was implemented successfully using C.